

lites were also put into low-earth orbits. The immediate problem with passive reflector at that time was extremely low signal level of the received signal. This required very costly high-power transmitters and highly sensitive receivers. The main reason for use of passive satellite was the lack of space-qualified electronics. It was recognised that amplifying the received signal onboard would greatly enhance the capability of the satellites for communication. Therefore considerable efforts were put in the development of space-qualified electronics, eventually leading to the introduction of active repeaters.

The first true telecommunication satellites, Telstar I and II, were launched in July 1962 and May 1963, respectively. The Telstar satellites were built by Bell Telephone Laboratories and used C-band transponders adopted from terrestrial microwave link equipment. The uplink was at 6389 Hz and the downlink was at 4169 MHz, with 50MHz bandwidth. The satellites carried solar cells and batteries that allowed continuous use of the single transponder. Live television links and multiplexed telephone circuits were established across the Atlantic Ocean and emphatically demonstrated the feasibility of satellite communication.

These two satellites were launched into what may be considered medium-earth orbit (MEO) with an orbit period of 158 minutes and 225 minutes, respectively. This allowed about 20-minute transatlantic links while the satellite was mutually visible. The orbits chosen for the Telstar satellites took them through high-energy radiation bands, which caused early failure of electronics onboard. However the value of satellite communication had been demonstrated.

In mid-1963, majority of all satellites had been launched into low-earth orbit (LEO). LEO and MEO were much easier to reach than GEO with the small launchers available at that time. The reliability of the launcher was a serious concern at that time. The new frontier was very risky with about



Telstar I satellite

one launch in four being successful. Therefore first six years of the so-called 'space age' belonged to launcher and payload development.

The first geostationary satellite that worked well was SYNCOM III which was built by Hughes Aircraft Company and launched on August 19, 1964. The satellite was used to telecast 1964 summer Olympics held in Tokyo to the US.

On August 31, 1962, president John F. Kennedy signed the communication satellite act with the goal of establishing a commercial communication satellite system in cooperation with other nations. On August 19, 1964, representatives of the first twelve countries signed an initial agreement to establish International Telecommunications Satellite Organisation (Intelsat). The United States was represented by a company called Comsat at the initial signing ceremony.

On April 6, 1965, Intelsat-I (Nick named 'Early Bird') was launched by

Delta D launcher. The satellite was built for Comsat by Hughes Aircraft Company based on their SYNCOM experience. The satellite carried two 6/4GHz transponders, each with 25MHz bandwidth. Commercial operations commenced between the US and Europe. Thus two decades after Clarke's landmark article in *Wireless World*, GEO satellite communication began.

The first four Intelsat series of satellites (Intelsat I through IV) were selected and procurement was managed by teams put in place under Comsat leadership. Subsequently, when the permanent management arrangements came into force for Intelsat in 1979, it was a smooth transition with most of the same teams coming under Intelsat.

Long-distance communication

Requirement for long-distance communication was the system driver for satellite communication during the early days. Canada was the first country to build a national telecommunication system using GEO satellites. Anik A1 was launched in 1972 two years before the US domestic satellite WESTAR 1 was launched in 1974.

However, the honour of first regional satellite communication system goes to Molniya of the former Soviet Union. The first satellite of this system was launched in April 1965—the same month as Intelsat I. Molniya satellites were launched into a highly elliptic orbit (HEO) that had an orbit period of twelve hours and orbit inclination of around 65 degrees. It may be noted that a satellite in Molniya orbit repeatedly crosses high-energy Van Allan Belt and this reduces the life of the satellite.

Another country that greatly benefited by GEO communication was Indonesia, which consists of more than 3000 islands spread over more than 1600 km. A terrestrial communication system was not economically feasible for it, while a single GEO satellite (PALAPA launched in 1976) allowed instant communications region-wide.